

CCS WITH TRUE COSTS

Assume unequal misclassification costs, i.e., $cost_{FN} \neq cost_{FP}$ and generalize error rate to **expected costs** (as function of π_+):

$$Costs(\pi_+) = (1 - \pi_+) \cdot FPR \cdot cost_{FP} + \pi_+ \cdot FNR \cdot cost_{FN}$$

Maximum of expected costs happens when

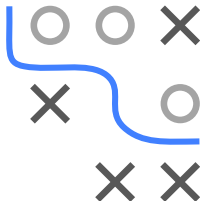
$$FPR = FNR = 1 \Rightarrow Costs_{max} = (1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}$$

Consider **normalized costs** (as function of π_+):

$$\begin{aligned} Costs_{norm}(\pi_+) &= \frac{(1 - \pi_+) \cdot FPR \cdot cost_{FP} + \pi_+ \cdot FNR \cdot cost_{FN}}{(1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}} \\ &= \frac{(1 - \pi_+) \cdot cost_{FP} \cdot FPR}{(1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}} + \frac{\pi_+ \cdot cost_{FN} \cdot FNR}{(1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}} \end{aligned}$$

Let "probability times cost" $PC(+)$ be normalized version of $\pi_+ \cdot cost_{FN}$:

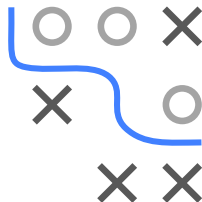
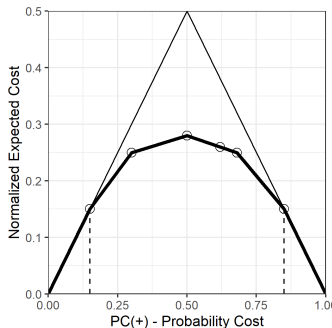
$$PC(+) = \frac{\pi_+ \cdot cost_{FN}}{(1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}} \text{ and } 1 - PC(+) = \frac{(1 - \pi_+) \cdot cost_{FP}}{(1 - \pi_+) \cdot cost_{FP} + \pi_+ \cdot cost_{FN}}$$



COMPARE WITH TRIVIAL CLASSIFIERS

- Operating range of a classifier is a set of $PC(+)$ values (operating points) where classifier performs better than both trivial classifiers
- Intersection of cost curves and trivial classifiers' diagonals determine operating range
- At any $PC(+)$ value, the vertical distance of trivial diagonal to a classifier's cost curve within operating range shows advantage in performance (normalized costs) of classifier

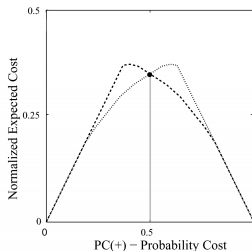
Example: Dotted lines are operating range of a classifier (here: $[0.14, 0.85]$)



COMPARING CLASSIFIERS

- If classifier C1's expected cost is lower than classifier C2's at a $PC(+)$ value, C1 outperforms C2 at that operating point
- The two cost curves of C1 and C2 may cross, which indicates C1 outperforms C2 for a certain operating range and vice versa
- The vertical distance between the two cost curves of C1 and C2 at any $PC(+)$ value directly indicates the performance difference between them at that operating point

Example: Dotted cost curve has lower expected cost as dashed cost curve for $PC(+) < 0.5$ and hence outperforms dashed one in this operating range and vice versa



Chris Drummond and Robert C. Holte (2006):
Cost curves: An improved method for visualizing
classifier performance. *Machine Learning*, 65,
95-130 (URL)

